

ENGINEERING CHEMISTRY — FORMULA SHEET

Surface Chemistry & Nanomaterials

B.Tech 1st Year · 1st Semester · Last-Day Revision Tool



UNIT V — Complete Revision Sheet

LEVEL

B.Tech 1st Yr

SEMESTER

Sem — I

PAGES

10–15 Pages

⚙ Surface Chemistry

📊 Adsorption Isotherms

🌿 Colloids

🔗 Micelle Formation

🧪 Nanomaterials

🔬 Synthesis Methods

🛡 Stabilization

🚀 Applications

Contains: Formulas · Definitions · Classifications · Mechanisms · Flashcards · Exam Tips

1 COURSE OBJECTIVES

- ✦ Understand fundamentals of surface chemistry
- ✦ Understand colloids and micelle formation
- ✦ Explore synthesis and stabilization methods
- ✦ Learn adsorption phenomena and isotherms
- ✦ Study nanometals and metal oxide nanomaterials
- ✦ Applications in medicine, energy, sensors, industry

2 SURFACE CHEMISTRY — INTRODUCTION

DEFINITION — SURFACE CHEMISTRY

Branch of chemistry that deals with phenomena occurring at the **interface (surface/boundary)** between two phases (solid-liquid, liquid-gas, solid-gas). Surfaces have higher energy than the bulk.

ADSORPTION

- ▶ Accumulation of molecules **on the surface**
- ▶ Surface phenomenon
- ▶ Adsorbate collects on adsorbent
- ▶ Example: Charcoal adsorbs gases
- ▶ Process is **exothermic**

ABSORPTION

- ▶ Uniform distribution **throughout bulk**
- ▶ Bulk phenomenon
- ▶ One substance enters another
- ▶ Example: Sponge absorbs water
- ▶ Can be exothermic or endothermic

Physisorption vs Chemisorption — Comparison Table

PROPERTY	PHYSISORPTION (PHYSICAL ADSORPTION)	CHEMISORPTION (CHEMICAL ADSORPTION)
Force	Weak van der Waals forces	Strong chemical bonds (covalent/ionic)
Heat of Adsorption	Low (~20–40 kJ/mol)	High (~80–400 kJ/mol)
Specificity	Non-specific	Highly specific
Reversibility	Reversible (easily desorbed)	Irreversible / difficult to desorb
Layers	Multilayer formation possible	Only monolayer formation
Temperature Effect	Decreases with increasing temperature	Initially increases, then decreases
Activation Energy	Not required	Required (activated adsorption)
Example	N ₂ on charcoal at low temperature	H ₂ on Ni surface

FACTORS AFFECTING ADSORPTION

- ✓ **Surface Area:** ↑ surface area → ↑ adsorption
- ✓ **Temperature:** Physisorption ↓ with ↑ T; Chemisorption ↑ then ↓
- ✓ **Pressure:** ↑ pressure → ↑ adsorption (gases)
- ✓ **Nature of adsorbate/adsorbent:** polarity, molecular size matter

3



ADSORPTION ISOTHERMS

WHAT IS AN ADSORPTION ISOTHERM?

A graph/equation showing the relationship between the **amount of adsorbate adsorbed per unit mass of adsorbent (x/m)** and the **pressure (p) or concentration (c) at constant temperature**.

① FREUNDLICH ADSORPTION ISOTHERM (1909) — EMPIRICAL

FREUNDLICH EQUATION

$$x/m = K \cdot P^{1/n}$$

x = mass of adsorbate | m = mass of adsorbent | P = pressure | K, n = Freundlich constants ($n > 1$)

LOGARITHMIC (LINEAR) FORM — USED TO PLOT GRAPH

$$\log(x/m) = \log K + (1/n) \cdot \log P$$

Plot: $\log(x/m)$ vs $\log P \rightarrow$ Straight line | Slope = $1/n$ | Intercept = $\log K$

- Valid for **non-ideal systems** and **heterogeneous surfaces**
- $1/n$: $0 < 1/n < 1$ for favourable adsorption
- Does NOT predict adsorption at high pressures (no saturation value)

② LANGMUIR ADSORPTION ISOTHERM (1916) — THEORETICAL

KEY ASSUMPTIONS

- Adsorption forms a **monomolecular layer** (monolayer) only
- All adsorption sites are **equivalent and identical**
- No lateral interaction between adsorbed molecules
- Adsorption is a **dynamic equilibrium** (adsorption rate = desorption rate)
- Surface is **homogeneous**

LANGMUIR EQUATION

$$x/m = a \cdot b \cdot P / (1 + b \cdot P)$$

a = monolayer capacity | b = Langmuir constant (adsorption affinity) | P = pressure

LINEARISED LANGMUIR FORM

$$P / (x/m) = 1/(a \cdot b) + P/a$$

Plot: $P/(x/m)$ vs $P \rightarrow$ Straight line | Slope = $1/a$ | Intercept = $1/(a \cdot b)$

③ BET EQUATION (BRUNAUER–EMMETT–TELLER, 1938) — MULTILAYER

BET EQUATION (FULL FORM)

$$P / [V(P_0 - P)] = 1 / (V_m \cdot C) + [(C - 1) / (V_m \cdot C)] \cdot (P / P_0)$$

P = pressure | P₀ = saturation pressure | V = volume adsorbed | V_m = monolayer volume | C = BET constant

- **Purpose:** Determination of **specific surface area** of adsorbent
- Extension of Langmuir to **multilayer adsorption**
- Plot: P/[V(P₀–P)] vs P/P₀ → Straight line
- Slope = (C–1)/(V_m·C) ; Intercept = 1/(V_m·C)
- Surface Area (S) = V_m × N_A × σ / 22414 (cm²/g)

 Freundlich vs Langmuir — Quick Comparison

PARAMETER	FREUNDLICH ISOTHERM	LANGMUIR ISOTHERM
Type	Empirical	Theoretical
Surface	Heterogeneous	Homogeneous
Layer	Multilayer	Monolayer only
Saturation	No definite saturation	Definite saturation value (a)
Equation	x/m = KP ^{1/n}	x/m = abP/(1+bP)
Linear Plot	log(x/m) vs log P	P/(x/m) vs P
Validity	Low to medium pressure	All pressures
Application	Gas adsorption on charcoal	Chemisorption, enzyme kinetics

4



COLLOIDS

DEFINITION — COLLOID / COLLOIDAL SYSTEM

A heterogeneous system in which particles of size **1 nm to 1000 nm (1 μm)** are dispersed in a dispersion medium. Colloidal particles do NOT settle under gravity and cannot be filtered through ordinary filter paper.

CLASSIFICATION BY AFFINITY FOR DISPERSION MEDIUM

LYOPHILIC COLLOIDS (SOLVENT-LOVING)

- ▶ Strong affinity between dispersed phase & medium
- ▶ Stable, reversible
- ▶ Formed easily by mixing
- ▶ Examples: gelatin, starch, gum, rubber (in water)

LYOPHOBIC COLLOIDS (SOLVENT-HATING)

- ▶ Weak/no affinity for dispersion medium
- ▶ Unstable, irreversible
- ▶ Require special preparation
- ▶ Examples: metal sols (Au, $\text{Fe}(\text{OH})_3$, As_2S_3)

Types of Colloids — Classification Table

TYPE	DISPERSED PHASE	DISPERSION MEDIUM	EXAMPLE
Multimolecular	Aggregates of small molecules (<1 nm each)	Liquid/gas	Gold sol, Sulphur sol
Macromolecular	Large single molecules (polymers)	Liquid	Starch, proteins, cellulose
Associated (Micelles)	Aggregates of amphiphilic molecules	Liquid	Soap, detergent solutions above CMC
Sol	Solid	Liquid	Paint, $\text{Fe}(\text{OH})_3$ sol
Gel	Liquid	Solid	Jelly, cheese
Aerosol	Solid/Liquid	Gas	Fog, smoke, cloud
Foam	Gas	Liquid/Solid	Shaving cream, foam rubber
Emulsion	Liquid	Liquid	Milk, mayonnaise

PROPERTIES OF COLLOIDS

TYNDALL EFFECT

Scattering of light by colloidal particles. The path of light becomes visible. Used to **distinguish colloids from true solutions**.

BROWNIAN MOTION

Random zigzag motion of colloidal particles due to **unequal bombardment** by medium molecules. Prevents sedimentation.

CHARGE ON PARTICLES

Colloidal particles carry **similar charge** (+ or -), causing mutual repulsion → stability. Measured by zeta potential (ζ).

5



MICELLE FORMATION

DEFINITION — MICELLE

Aggregates of **amphiphilic (surfactant) molecules** having both hydrophilic (head) and hydrophobic (tail) parts. Above CMC, molecules self-assemble into micelles in aqueous solution.

CRITICAL MICELLE CONCENTRATION (CMC)

- ▶ Minimum concentration at which micelles form
- ▶ Below CMC → surfactant molecules exist individually
- ▶ Above CMC → micelles form spontaneously
- ▶ CMC of soap (sodium stearate): ~1 mM
- ▶ Detected by sudden change in: conductivity, surface tension, turbidity

MICELLE STRUCTURE

- ▶ **Head:** Hydrophilic (ionic/polar) — faces outward toward water
- ▶ **Tail:** Hydrophobic (long carbon chain) — faces inward
- ▶ **Core:** Non-polar interior (can dissolve oils)
- ▶ **Shape:** Spherical (most common), cylindrical, bilayer

EXAMPLES OF MICELLE FORMING AGENTS

- ▶ Soap: $\text{CH}_3(\text{CH}_2)_{16}\text{COO}^-\text{Na}^+$ (sodium stearate)
- ▶ Detergents: SDS (sodium dodecyl sulphate)
- ▶ Bile salts (biological system)

6



SYNTHESIS OF COLLOIDS

Methods of Colloid Synthesis

METHOD	TYPE	KEY STEPS / PRINCIPLE	EXAMPLE
Bredig's Arc Method	Dispersion	Electric arc between metal electrodes in water → vapourisation + condensation of metal atoms	Gold, Silver, Platinum sols
Chemical Reduction	Condensation	Metal salt + reducing agent → metal nanoparticles	$\text{HAuCl}_4 + \text{citrate} \rightarrow \text{Au nanoparticles}$
Peptization	Dispersion	Adding small amount of electrolyte (peptizing agent) to freshly precipitated colloid	$\text{Fe}(\text{OH})_3 \text{ precipitate} + \text{FeCl}_3 \rightarrow \text{Fe}(\text{OH})_3 \text{ sol}$
Double Decomposition	Condensation	Reaction of two soluble salts → insoluble colloidal product	$\text{H}_2\text{S} + \text{As}_2\text{O}_3 \rightarrow \text{As}_2\text{S}_3 \text{ sol}$
Oxidation/Reduction	Condensation	Controlled chemical reaction to generate colloidal particles	$\text{SO}_2 + \text{H}_2\text{S} \rightarrow \text{S sol}$

7



STABILIZATION OF COLLOIDS

ELECTRICAL DOUBLE LAYER (EDL / STERN LAYER)

Colloidal particles develop a charge → attracts opposite ions → forms an inner Stern layer and outer diffuse layer. Repulsion between similarly charged particles → **electrostatic stabilization**.

STABILIZING AGENTS

- ▶ **Protective colloids:** Gelatin, casein (steric stabilization)
- ▶ **Surfactants:** Coat particles, prevent aggregation
- ▶ **Electrolytes:** Maintain charge on particles
- ▶ **Gold Number:** minimum amount of protective colloid to prevent coagulation of 10 mL gold sol

APPLICATIONS OF COLLOIDS

- ✓ **Water purification:** Alum coagulates colloidal impurities
- ✓ **Medicines:** Colloidal silver (antiseptic), argyrol
- ✓ **Food industry:** Milk, butter, ice cream (emulsions)
- ✓ **Photography:** AgBr colloidal suspension on film
- ✓ **Rubber industry:** Latex (natural rubber sol)
- ✓ **Industrial:** Electrophoresis, dialysis, smoke precipitation

8



NANOMATERIALS

DEFINITION — NANOMATERIALS

Materials with at least one dimension in the range of **1–100 nanometres (nm)**. At nanoscale, materials exhibit unique physical, chemical, optical, and electronic properties different from their bulk counterparts.

SIZE REFERENCE

- 1 nm = 10^{-9} m
- Nanomaterial range: **1–100 nm**
- Human hair width: ~80,000 nm
- DNA helix width: ~2 nm
- Red blood cell: ~8000 nm

Types of Nanomaterials

DIMENSIONALITY	NAME	DESCRIPTION	EXAMPLE
0D	Quantum Dots / NPs	All dimensions at nanoscale	Gold NPs, Carbon dots
1D	Nanowires / Nanorods	One dimension extended	Carbon nanotubes, ZnO nanorods
2D	Nanosheets / Thin films	Two dimensions extended	Graphene, MoS ₂ sheets
3D	Bulk nanostructures	All dimensions > 100 nm but nanostructured	Nanocomposites, nanoporous materials

TYPES: NANOMETALS VS NANOMETAL OXIDES

NANOMETALS

- Pure metal nanoparticles (0-valent)
- Examples: Au (Gold), Ag (Silver), Pt, Cu, Fe NPs
- Properties: Plasmon resonance (Au → red), high conductivity
- Applications: catalysis, medicine, sensors, electronics
- Synthesis: chemical reduction of metal salts

NANOMETAL OXIDES

- Oxide form of metals at nanoscale
- Examples: TiO₂, ZnO, Fe₂O₃, CuO, Al₂O₃, CeO₂
- Properties: photocatalytic, magnetic, semiconductor behaviour
- Applications: photocatalysis, UV protection, sensors, energy
- Synthesis: precipitation, sol-gel, hydrothermal methods

9



UNIQUE PROPERTIES OF NANOMATERIALS

PROPERTY	DESCRIPTION	REASON/MECHANISM
High Surface Area	Surface area-to-volume ratio increases drastically at nanoscale	More atoms at surface → higher reactivity and adsorption capacity
Quantum Confinement	Electronic and optical properties change with size	Electron motion confined → discrete energy levels → size-tunable properties
Surface Plasmon Resonance	Gold NPs appear red/purple instead of yellow	Collective oscillation of conduction electrons on absorption of light
Enhanced Reactivity	Higher catalytic and chemical reactivity	Large fraction of atoms exposed at surface; lower activation energy
Magnetic Properties	Superparamagnetism in Fe ₃ O ₄ NPs	Single magnetic domain; random thermal fluctuations dominate
Mechanical Strength	Higher strength than bulk material	Fewer defects/dislocations at nanoscale
Melting Point Reduction	Nanoparticles melt at lower temperatures	High surface energy; weaker average atomic bonding

10



PREPARATION / SYNTHESIS OF NANOMATERIALS

TOP-DOWN APPROACH

- ▶ Breaking bulk material → nanoscale
- ▶ Methods: ball milling, lithography, etching, laser ablation
- ▶ Less control over size/shape
- ▶ Suitable for large-scale production

BOTTOM-UP APPROACH

- ▶ Building nanostructures atom by atom / molecule by molecule
- ▶ Methods: chemical synthesis, self-assembly, CVD
- ▶ Better control over size and shape
- ▶ Higher purity products

Synthesis Methods — Comparison Table

METHOD	TYPE	PRINCIPLE	EXAMPLE / PRODUCT	ADVANTAGE
Chemical Reduction	Chemical / Bottom-up	Metal salt + reducing agent → metal NPs e.g. $\text{HAuCl}_4 + \text{NaBH}_4 \rightarrow \text{Au NPs}$	Au, Ag, Pt NPs	Simple, scalable, size-tunable
Co-precipitation	Chemical / Bottom-up	Mixing metal salts + precipitating agent (NaOH , NH_3) → metal oxide NPs	Fe_3O_4 , ZnO, CuO NPs	Simple, room temperature possible
Sol-Gel Method	Chemical / Bottom-up	Metal alkoxide hydrolysis → gel → calcination → nanoparticles	SiO_2 , TiO_2 , Al_2O_3	High purity, controlled porosity
Hydrothermal	Chemical / Bottom-up	High T and P water (autoclave) → crystal growth of metal oxides	ZnO, TiO_2 nanorods	Crystalline, controlled morphology
Green Synthesis (Plant)	Biological / Bottom-up	Plant extract (phytochemicals) acts as reducing + capping agent	Ag, Au, ZnO NPs	Eco-friendly, no toxic chemicals
Microbial Synthesis	Biological / Bottom-up	Bacteria/fungi reduce metal ions extracellularly or intracellularly	Ag NPs (E. coli, Fusarium)	Biocompatible, sustainable
Ball Milling	Physical / Top-down	Mechanical grinding of bulk material → fine nanoparticles	Metal/oxide powders	Large scale, no solvents

GREEN SYNTHESIS — KEY POINTS

- ▶ Uses plant extracts: neem, tulsi, turmeric, aloe vera
- ▶ Phytochemicals (flavonoids, terpenoids) act as **reducing + capping agents**
- ▶ No harsh chemicals, no high temperature needed
- ▶ Biocompatible NPs — suitable for biomedical applications
- ▶ Common: AgNO_3 + plant extract → Ag NPs (colour change: yellow → brown)

MICROBIAL / BIOLOGICAL SYNTHESIS

- ▶ Bacteria, fungi, algae, yeast used
- ▶ Extracellular: NPs formed outside the cell
- ▶ Intracellular: NPs formed inside the cell
- ▶ Example: *Trichoderma* spp. → Ag NPs
- ▶ Advantage: Large scale, economical, biocompatible

11



STABILIZATION OF NANOMATERIALS

Stabilization Techniques — Summary Table

Stabilization Method	Agent / Material	Mechanism	Example
Steric Stabilization	Polymers (PVP, PEG)	Polymer chains adsorb on NP surface → steric repulsion prevents aggregation	PVP-capped Ag NPs
Electrostatic Stabilization	Charged capping agents, citrate	Surface charge → electrostatic repulsion → prevents aggregation	Citrate-capped Au NPs
Capping Agents	Thiols (–SH), amines, carboxylates	Bind to NP surface → passivation → prevent oxidation and aggregation	Thiol-capped Au NPs
Surfactant Stabilization	CTAB, SDS, Tween-80	Surfactant bilayer around NPs → steric + electrostatic protection	CTAB-Au nanorods
Ligand Exchange	Functional organic ligands	Replace weakly bound stabilizers with stronger, functional ligands	PEG-coated NPs for drug delivery



APPLICATIONS OF NANOMATERIALS



CATALYSIS

- ▶ High surface area → enhanced catalytic activity
- ▶ TiO_2 NPs: photocatalytic degradation of pollutants
- ▶ Pt, Pd NPs: automotive catalytic converters
- ▶ Fe NPs: Fenton reaction for wastewater treatment
- ▶ Au NPs: CO oxidation at room temperature



MEDICINE & DRUG DELIVERY

- ▶ Targeted drug delivery (cancer therapy)
- ▶ Iron oxide NPs: MRI contrast agents
- ▶ Ag NPs: broad-spectrum antimicrobial agent
- ▶ Au NPs: photothermal cancer therapy
- ▶ Liposome NPs: encapsulate drugs for controlled release



SENSORS

- ▶ Au NPs: colorimetric biosensors (change colour on binding)
- ▶ ZnO NPs: gas sensors (detect NH_3 , NO_2 , H_2)
- ▶ Carbon nanotubes: chemical/biological sensors
- ▶ Surface-enhanced Raman scattering (SERS) sensors



ELECTRONICS

- ▶ Quantum dots: next-gen displays (QLED TV)
- ▶ Carbon nanotubes: transistors, nanowires
- ▶ Ag NPs: conductive inks for printed electronics
- ▶ Cu NPs: heat sinks, conductive pastes



ENERGY

- ▶ TiO_2 NPs: dye-sensitized solar cells (DSSC)
- ▶ Pt NPs on carbon: fuel cell electrodes
- ▶ Fe_2O_3 NPs: lithium-ion battery anodes
- ▶ Carbon nanotubes: supercapacitors
- ▶ ZnO nanorods: piezoelectric nanogenerators



ENVIRONMENT

- ▶ TiO_2 : photocatalytic air/water purification
- ▶ Fe NPs: in-situ remediation of contaminated groundwater
- ▶ Ag NPs: water disinfection
- ▶ Nano-membranes: desalination, water filtration
- ▶ ZnO: sunscreen (UV protection)

NANOMATERIAL	KEY PROPERTY	MAIN APPLICATION
Au (Gold) NPs	Surface Plasmon Resonance, biocompatible	Biosensors, cancer therapy, drug delivery
Ag (Silver) NPs	Antimicrobial, conductive	Wound healing, medical devices, inks
TiO_2 NPs	Photocatalytic, UV-active	Solar cells, water treatment, sunscreens
ZnO NPs	Semiconductor, piezoelectric	Gas sensors, UV filters, antibacterial
Fe_3O_4 NPs	Superparamagnetic	MRI agents, targeted drug delivery, hyperthermia
CuO NPs	Antimicrobial, semiconducting	Catalysis, solar energy, lubricants
Carbon Nanotubes	High strength, conductivity	Electronics, composites, drug delivery

13



FLASHCARDS — EXAM RAPID RECALL (30 CARDS)

Q1 What is Adsorption?

Accumulation of molecules/atoms on the surface of a substance (adsorbent), not throughout the bulk

Q3 What is Chemisorption?

Adsorption involving strong chemical bond formation; specific, irreversible, high enthalpy (~80–400 kJ/mol)

Q5 Key Langmuir assumption?

Monolayer adsorption; homogeneous surface; no lateral interactions; dynamic equilibrium

Q7 BET equation used for?

Determination of specific surface area of porous materials/adsorbents (multilayer adsorption model)

Q9 Tyndall Effect?

Scattering of light by colloidal particles making the beam visible; absent in true solutions

Q11 Lyophilic vs Lyophobic?

Lyophilic = solvent-loving, stable, reversible (gelatin). Lyophobic = solvent-hating, unstable, irreversible (Au sol)

Q13 Micelle structure?

Hydrophilic heads face outward (water), hydrophobic tails face inward (core); formed by amphiphilic molecules

Q15 Gold Number?

Minimum mass (mg) of protective colloid needed to prevent coagulation of 10 mL standard gold sol

Q17 Why do NPs have unique properties?

High surface area-to-volume ratio + quantum confinement effects at nanoscale → new electrical, optical, magnetic behaviour

Q19 Chemical reduction synthesis?

Metal salt + reducing agent (NaBH_4 , citrate) → metal NPs. Example: HAuCl_4 + citrate → Au NPs

Q21 PVP capping agent role?

Polyvinylpyrrolidone adsorbs on NP surface → steric repulsion → prevents NP aggregation

Q2 Physisorption uses which forces?

Weak van der Waals forces (reversible; low heat of adsorption ~20–40 kJ/mol)

Q4 Freundlich Isotherm formula?

$$x/m = K \cdot P^{1/n} \rightarrow \log(x/m) = \log K + (1/n) \log P$$

Q6 Langmuir equation?

$$x/m = a \cdot b \cdot P / (1 + b \cdot P); \text{Linear: } P/(x/m) = 1/(ab) + P/a$$

Q8 What is a Colloid?

Heterogeneous system with dispersed phase particle size 1–1000 nm in a dispersion medium

Q10 Brownian Motion?

Random zigzag movement of colloidal particles due to unequal bombardment by medium molecules

Q12 What is CMC?

Critical Micelle Concentration — minimum concentration at which surfactant molecules form micelles

Q14 Bredig's Arc Method?

Electric arc between metal electrodes submerged in water → metal vapour condenses → metal colloid

Q16 Nanomaterial size range?

1 nm to 100 nm (at least one dimension in this range)

Q18 Surface Plasmon Resonance?

Collective oscillation of conduction electrons in metallic NPs on light absorption — Au NPs appear red/purple

Q20 Green synthesis principle?

Plant phytochemicals (polyphenols, flavonoids) reduce metal ions AND cap NPs — eco-friendly, no toxic agents

Q22 Superparamagnetism?

Magnetic NPs (Fe_3O_4) with single domain behave like paramagnets with high susceptibility — used in MRI agents

Q23 **TiO₂ NP application?**

Photocatalysis (water/air purification), solar cells (DSSC), sunscreens (UV blocking)

Q25 **Freundlich graph: what is the slope?**

Slope of $\log(x/m)$ vs $\log P = 1/n$ (Freundlich constant, related to adsorption intensity)

Q24 **Ag NP main property & use?**

Broad-spectrum antimicrobial; used in wound dressings, medical devices, water purification

Q26 **Co-precipitation method?**

Mix metal salt solutions + precipitating agent (NaOH/NH₃) → simultaneous precipitation → metal oxide NPs

Q27 **Electrical Double Layer?**

Charged layer on colloidal particle (Stern layer + diffuse layer) → electrostatic repulsion → prevents coagulation

Q28 **Peptization?**

Converting a fresh precipitate into a stable colloid by adding a small amount of electrolyte (peptizing agent)

Q29 **What is dialysis?**

Purification of colloid by removing dissolved electrolytes using a semi-permeable membrane (colloid cannot pass)

Q30 **Quantum Confinement?**

When particle size ~ de Broglie wavelength of electron, energy levels become discrete → size-dependent optical/electrical properties

14

**EXAM TIPS — COMMON MISTAKES & HOW TO AVOID THEM**

- ✗ **Confusing Adsorption vs Absorption:** Adsorption = *surface phenomenon* (molecules stick ON the surface). Absorption = *bulk phenomenon* (one substance enters another entirely). Remember: Ad-Surface-orption. The "d" in adsorption = surface!
- ✗ **Mixing Freundlich and Langmuir Equations:** Freundlich: $x/m = KP^{1/n}$ (empirical, no derivation, heterogeneous surface). Langmuir: $x/m = abP/(1+bP)$ (theoretical, monolayer, homogeneous). Never swap the two equations in answers!
- ✗ **Forgetting Langmuir Assumptions:** Always state ALL 4 — (1) Monolayer only, (2) Homogeneous surface, (3) No lateral interactions between adsorbate molecules, (4) Dynamic equilibrium exists. Missing any one costs marks.
- ✗ **BET Equation Errors:** BET is for **multilayer adsorption** and **surface area determination** — not for studying basic adsorption. Remember variables: P_0 = saturation pressure, V_m = monolayer capacity. Don't confuse BET with Langmuir (monolayer only).
- ✗ **Misunderstanding Micelle Formation:** Micelles only form **ABOVE CMC**. Below CMC, individual surfactant molecules exist. The interior of micelle is **hydrophobic** (oily), exterior is **hydrophilic**. This is why soap cleans oil in water.
- ✗ **Mixing Chemical vs Biological Synthesis:** Chemical reduction uses NaBH_4 , citrate, hydrazine (lab chemicals). Green/Biological synthesis uses plant extract or microbes (no external reducing agent needed — phytochemicals do the reduction AND capping). Both are bottom-up approaches.
- ✗ **Ignoring Stabilization Mechanisms:** Always mention TWO types — (1) Electrostatic stabilization (surface charge/zeta potential) and (2) Steric stabilization (polymer/capping agents). Stabilization prevents aggregation and maintains nanoparticle properties.
- ✗ **Lyophilic vs Lyophobic Confusion:** Lyophilic = stable, reversible, no need for special preparation (gelatin, starch). Lyophobic = unstable, irreversible, need special preparation (metal sols). LYOPHILIC colloids protect LYOPHOBIC ones.
- ✗ **Nanomaterial Size Errors:** Size range = **1–100 nm**. Don't write 1–1000 nm (that's colloid range). Don't write micrometres (μm). $1 \text{ nm} = 10^{-9} \text{ m}$ — memorize this unit conversion.
- ✓ **Pro Tip — Freundlich Graph:** When plotting $\log(x/m)$ vs $\log P$ — Slope = $1/n$, Y-intercept = $\log K$. If $1/n < 1$, favourable adsorption. If $1/n = 1$, linear isotherm. If $1/n > 1$, cooperative adsorption (rare).
- ✓ **Pro Tip — Temperature & Adsorption:** For Physisorption: adsorption always DECREASES with increasing temperature (exothermic). For Chemisorption: adsorption initially INCREASES (activation needed), then DECREASES at very high temperatures.
- ✓ **Pro Tip — Tyndall vs True Solution:** If a beam of light becomes visible = Tyndall effect = colloidal solution. If beam invisible = true solution (particles too small to scatter). Use this to distinguish in MCQ!

15



QUICK REVISION SHEET — ALL-IN-ONE REFERENCE

KEY DEFINITIONS

TERM	ONE-LINE DEFINITION
Surface Chemistry	Study of phenomena at the interface of two phases
Adsorption	Accumulation of adsorbate on adsorbent surface
Absorption	Uniform distribution of substance throughout the bulk
Adsorbent	Substance on whose surface adsorption occurs (solid)
Adsorbate	Substance which gets adsorbed on the surface
Adsorption Isotherm	Plot of x/m vs P or C at constant temperature
Colloid	Heterogeneous system; particle size 1–1000 nm; doesn't settle
Sol	Solid dispersed in liquid
Gel	Liquid dispersed in solid
Emulsion	Liquid dispersed in liquid
Tyndall Effect	Scattering of light by colloidal particles
Brownian Motion	Random motion of colloidal particles due to molecular bombardment
Micelle	Aggregate of surfactant molecules (hydrophilic outside, hydrophobic core)
CMC	Minimum concentration for micelle formation
Gold Number	Min. mg of colloid to protect 10 mL gold sol from coagulation
Peptization	Converting fresh precipitate to colloid using peptizing agent
Dialysis	Purification of colloid using semi-permeable membrane
Nanomaterial	Material with at least one dimension in 1–100 nm range
Quantum Confinement	Discrete energy levels due to electron restriction at nanoscale
Surface Plasmon Resonance	Collective electron oscillation in metal NPs on light absorption
Green Synthesis	Eco-friendly NP synthesis using plant/microbial agents
Capping Agent	Molecules that bind to NP surface to prevent aggregation

ALL KEY FORMULAS AT A GLANCE

FREUNDLICH ISOTHERM

$$x/m = K \cdot P^{1/n}$$

LANGMUIR — LINEAR FORM

$$P/(x/m) = 1/(a \cdot b) + P/a$$

FREUNDLICH — LOG FORM

$$\log(x/m) = \log K + (1/n) \cdot \log P$$

BET EQUATION

$$P/[V(P_o - P)] = 1/(V_m C) + (C - 1)P/(V_m C \cdot P_o)$$

LANGMUIR ISOTHERM

$$x/m = a \cdot b \cdot P / (1 + b \cdot P)$$

SURFACE AREA (BET)

$$S = V_m \times N_A \times \sigma / 22414$$

KEY CONCEPTS — MEMORY TAGS

Physisorption: van der Waals

Chemisorption: Chemical Bond

Freundlich: Empirical

Langmuir: Monolayer

BET: Surface Area

Colloid: 1–1000 nm

Tyndall: Light Scattering

Brownian: Random Motion

Micelle: Above CMC

Lyophilic: Stable

Lyophobic: Unstable

Nanomaterial: 1–100 nm

Green Synthesis: Phytochemicals

SPR: Electron Oscillation

Quantum Confinement: Size Effect

Capping Agent: Stabilizer

Au NPs: Red/Purple colour

Ag NPs: Antimicrobial

TiO₂: Photocatalysis

Fe₃O₄: MRI Agent

ZnO: Gas Sensor

Dialysis: Purification

Gold Number: Protection

CLASSIFICATION SUMMARY

COLLOIDS BY TYPE

Sol (Solid/Liq) · **Emulsion** (Liq/Liq) · **Foam** (Gas/Liq) · **Aerosol** (Solid or Liq/Gas)
· **Gel** (Liq/Solid) · **Solid Foam** (Gas/Solid)

NANOMATERIALS BY DIMENSION

0D: Quantum dots, NPs · **1D**: Nanowires, CNTs · **2D**: Graphene, nanosheets ·
3D: Nanocomposites

IMPORTANT NUMBERS TO REMEMBER

NANOMATERIAL SIZE

1 – 100 nm

COLLOID PARTICLE SIZE

1 – 1000 nm

1 NM =

10⁻⁹ m

PHYSISORPTION HEAT

20–40 kJ/mol

CHEMISORPTION HEAT

80–400 kJ/mol

CMC OF SOAP

~1 mM

LAST-MINUTE REMINDERS

- ✓ Adsorption is always **exothermic**
- ✓ Freundlich = Empirical; Langmuir = Theoretical
- ✓ BET = Multilayer = Surface Area measurement
- ✓ Micelles form only **above** CMC
- ✓ Lyophilic = stable; Lyophobic = unstable
- ✓ Nanomaterials: 1–100 nm (NOT micrometres)
- ✓ Green synthesis: plant = reducing + capping agent
- ✓ Surface Plasmon Resonance makes Au NPs appear red